

1. Set up an integral to find the area of the region bounded by $y = 2x^2$ and $y = x^3$.
2. Set up an integral to find the volume of the solid generated when the region bounded by $y = 2x^2$ and $y = x^3$ is
 - (a) Rotated about the x -axis using shells
 - (b) Rotated about the x -axis using washers
 - (c) Rotated about the y -axis using shells
 - (d) Rotated about the y -axis using washers
 - (e) Rotated about the line $x = -3$
 - (f) Rotated about the line $y = -2$
 - (g) Rotated about the line $y = 11$
 - (h) Rotated about the line $x = 7$
3. Suppose a solid has the region bounded by $y = 2x^2$ and $y = x^3$ as a base. Set up integrals to find the volume if the cross-sections are
 - (a) Circles
 - (b) Semi-circles
 - (c) Rectangles in which the height is 3 times the base
4. Graph $r = \cos \theta$.
5. Graph $r = 5(2 - \cos \theta)$.